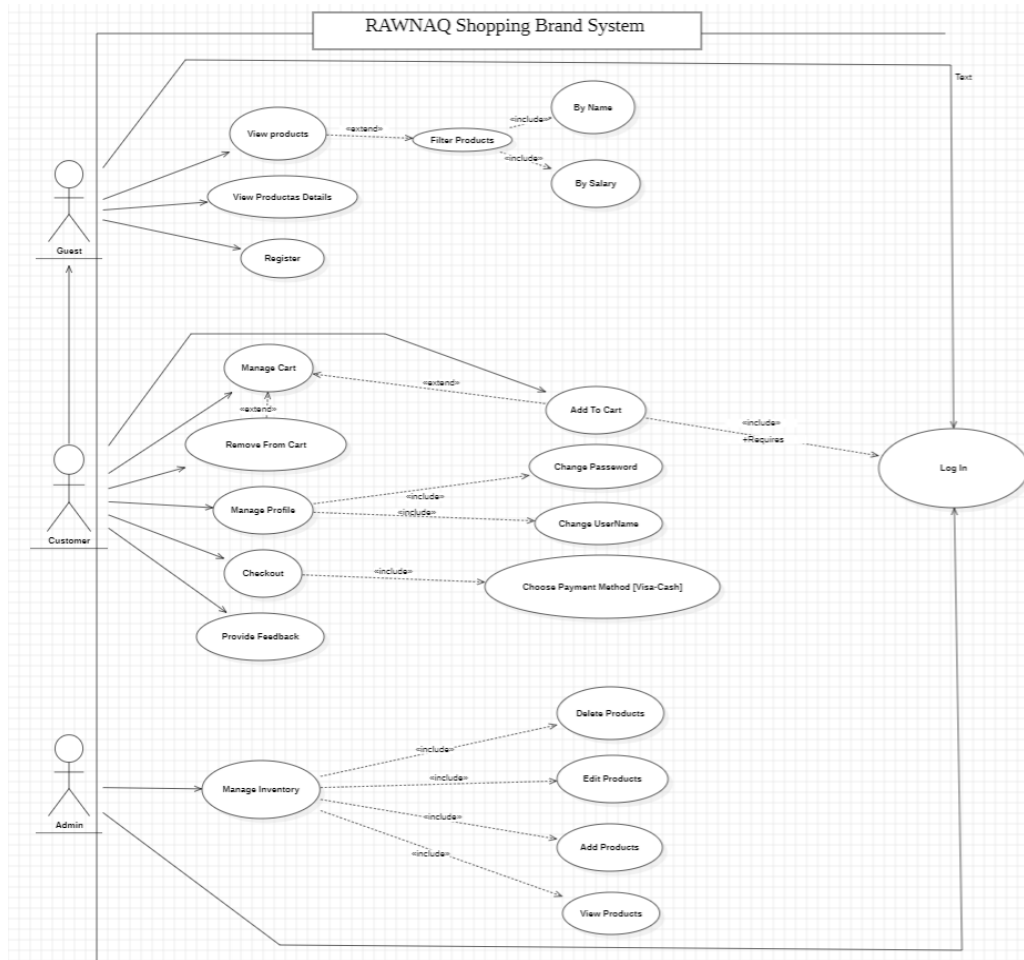


Figure 1 illustrates the main interactions between the users and the RAWNAQ E-Commerce System. The system involves three primary actors:

- **Guest:** A user who is not logged in. They can only **Browse Products, Search, and Filter** items by category or price.
- **Registered User (Client):** A user who has logged in. In addition to browsing, they can **Add Items to Cart, Manage Profile, Update Password, and perform Checkout** to complete a purchase.
- **Admin:** A user with special privileges who accesses the Dashboard to **Add Products, Edit Product Details, Delete Products, and Monitor Inventory.**
- The diagram below depicts these relationships and the system boundaries



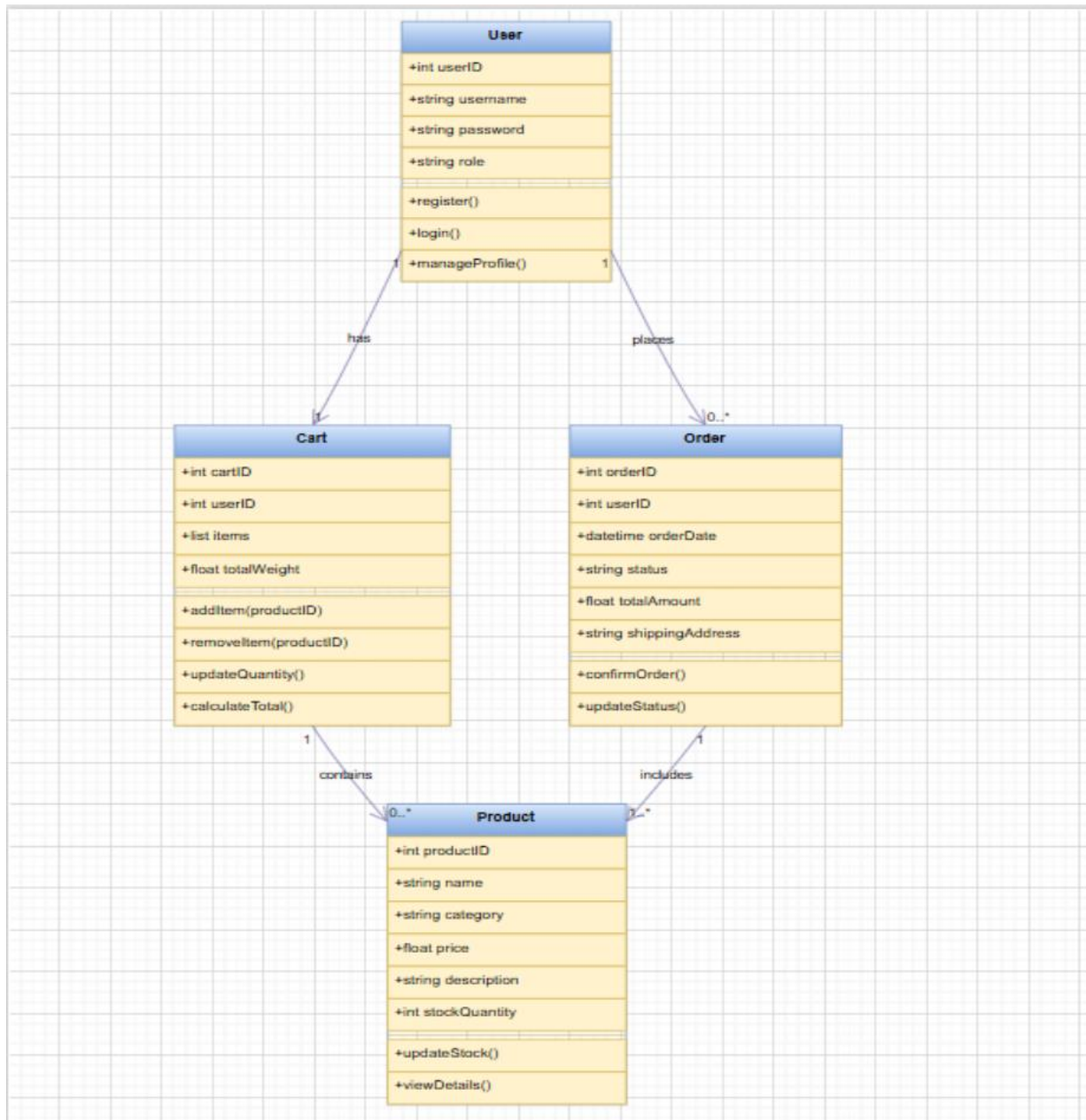


Figure 2 depicts the static structure of the RAWNAQ system. The main classes representing the database entities are:

- **User:** Represents the registered client or admin. It holds attributes like username, password, and role. It is associated with the **Cart** class.
- **Product:** Represents the fashion items available for sale. It contains details such as name, salary (price), stock, size, and image. Each product belongs to a specific **Category**.

- **Category:** Represents the classification of products (e.g., Men, Women). It has a one-to-many relationship with the **Product** class.
- **Cart:** Acts as a linking entity between **User** and **Product**. It stores the quantity of items a user intends to purchase.
- **Admin:** A specialized type of User (inherits from User) responsible for managing the inventory (CRUD operations).

3. Behavioral Model and Description

3.1. Description for Software Behavior

- **Startup:** When the application starts, it initializes the database (init_db.py) and ensures the default Admin user exists.
- **Navigation:** Users start at the Home page. If they try to access Admin pages without the 'admin' role in session_state, the system blocks access and shows a warning.
- **Transaction:** When a payment is successful, the behavioral logic ensures that the Stock is reduced instantly to prevent overselling, and the Cart is emptied.

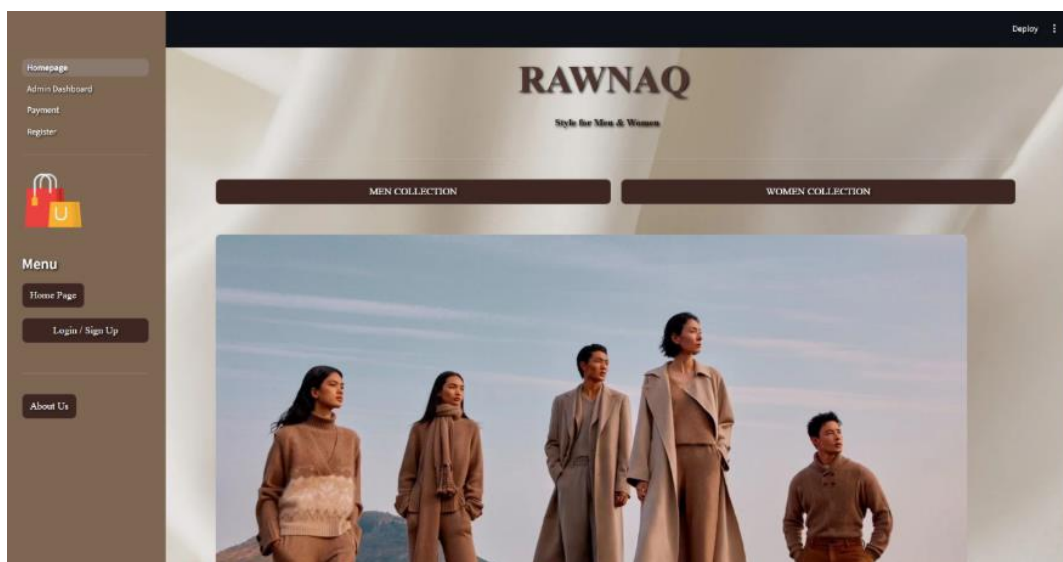


Figure 3. Picture of Main Page

Users can display the login page via “Login” button. The users can enter the system by entering username and password on this page.

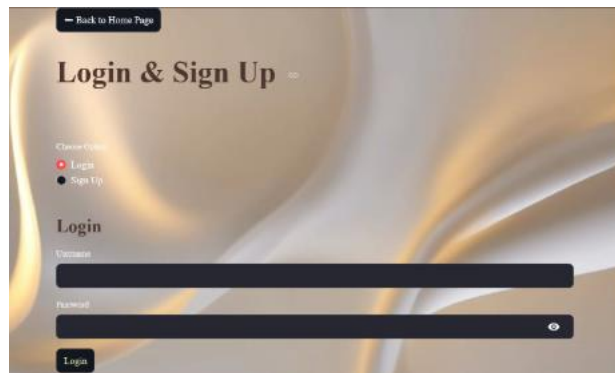
A screenshot of a web application's login and sign-up page. The page has a light beige background with a subtle, abstract pattern. At the top left, there is a dark button labeled "Back to Home Page". Below it, the title "Login & Sign Up" is displayed in a large, bold, dark font. Under the title, there are two radio buttons: "Login" (selected) and "Sign Up". Below the radio buttons, the "Login" section is visible, featuring a "Username" label, a dark input field, a "Password" label, another dark input field with a toggle icon on the right, and a dark "Login" button at the bottom.

Figure 4. Picture of log In Page

If they do not have an account, they can register the system by clicking “Sign Up” button. In this page, users have to fill the form which is about their username, password

A screenshot of the same web application's login and sign-up page, but with the "Sign Up" section selected. The "Sign Up" radio button is now selected, and the "Sign Up" section is visible. It features a "New Username" label, a dark input field, a "New Password" label, another dark input field with a toggle icon on the right, and a dark "Sign Up" button at the bottom. The "Login" section is no longer visible.

Figure 5. Visualization of Sign Up Page

When they enter the system, they see their profile information such as name, password, and they can change them, and log out

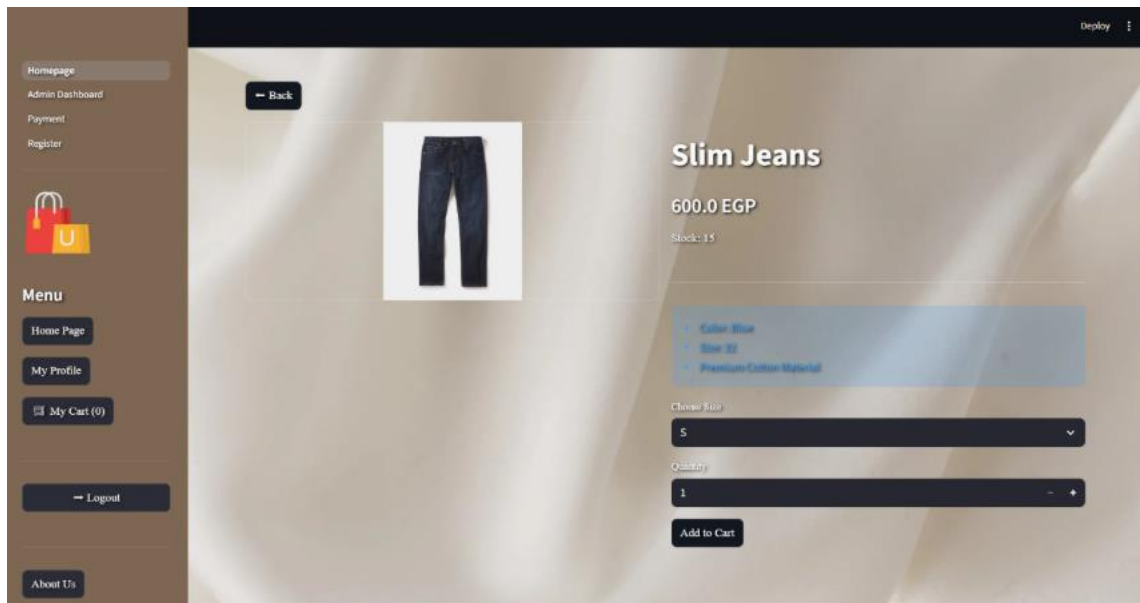


Figure 6. Visualization of Profile (user) Page

?

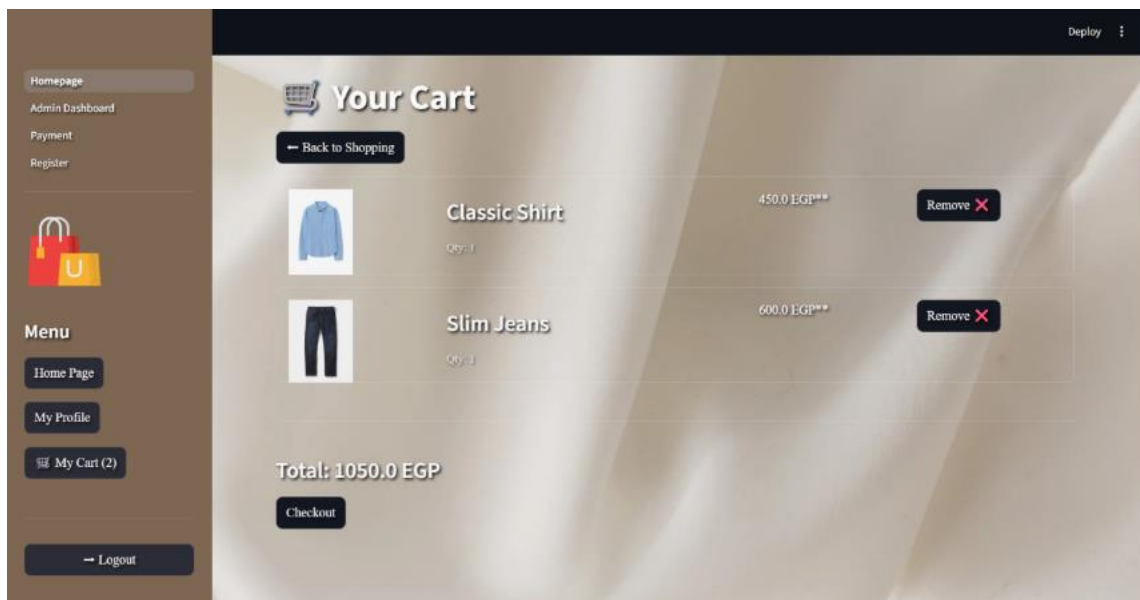
Figure 7. Visualization of Products & Filtering Page

User can Chose products and add it to the cart and filter the products by price and name.



Then if the user click on product, the page of details will shown.

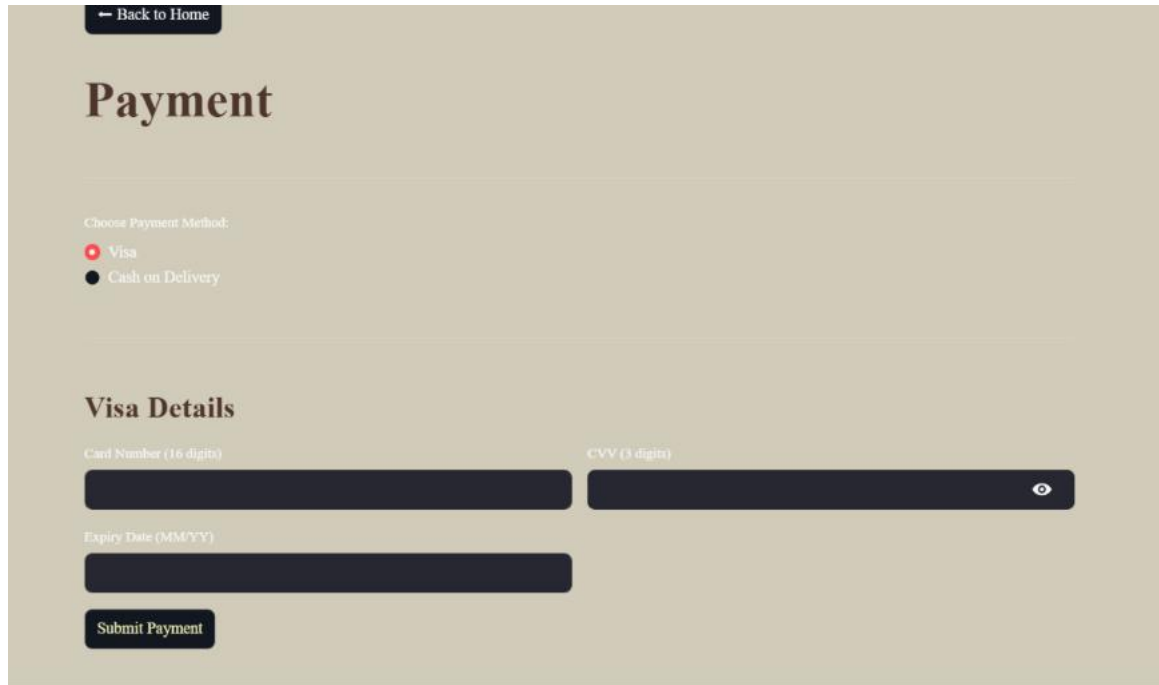
Figure 8. Visualization of Details Page



Moreover, users can see his cart and remove products from it and move to checkout.

Figure 9. Visualization of Cart Page

If user go to the checkout he will go to the Payment page.



← Back to Home

Payment

Choose Payment Method:

☒ Visa

☐ Cash on Delivery

Visa Details

Card Number (16 digits)

CVV (3 digits)

Expiry Date (MM/YY)

Submit Payment

Figure

10. Visualization of Payment Page

If user log in as Admin he will move to the Admin Dashboard page.

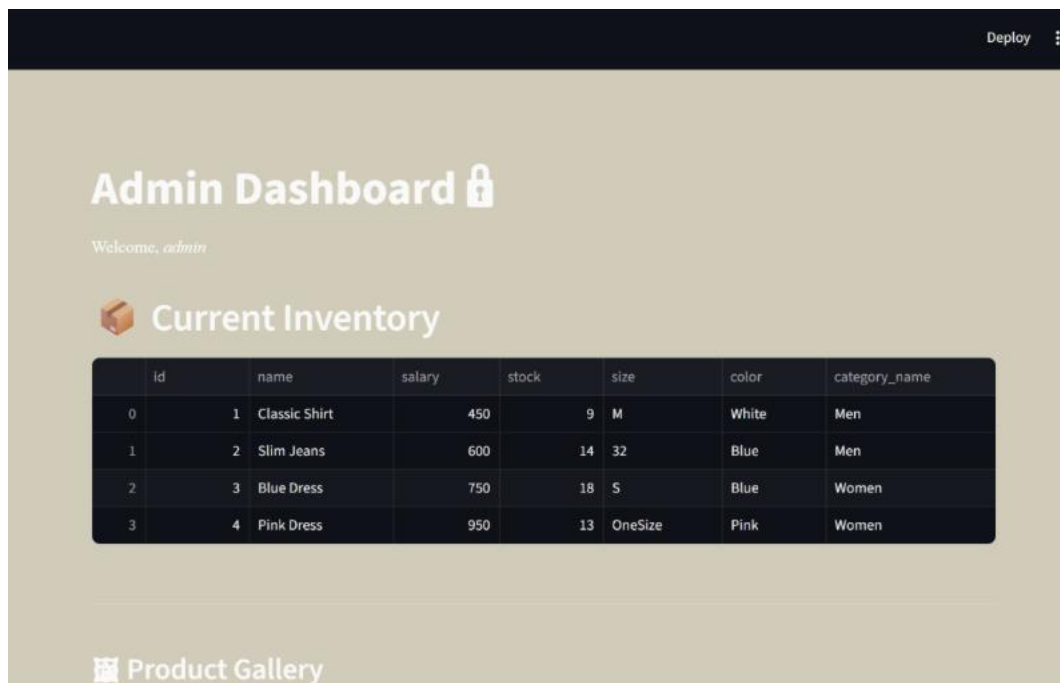


Figure 11. Visualization of Admin Dashboard Page

When click add product the Add product page will open, and he can add.

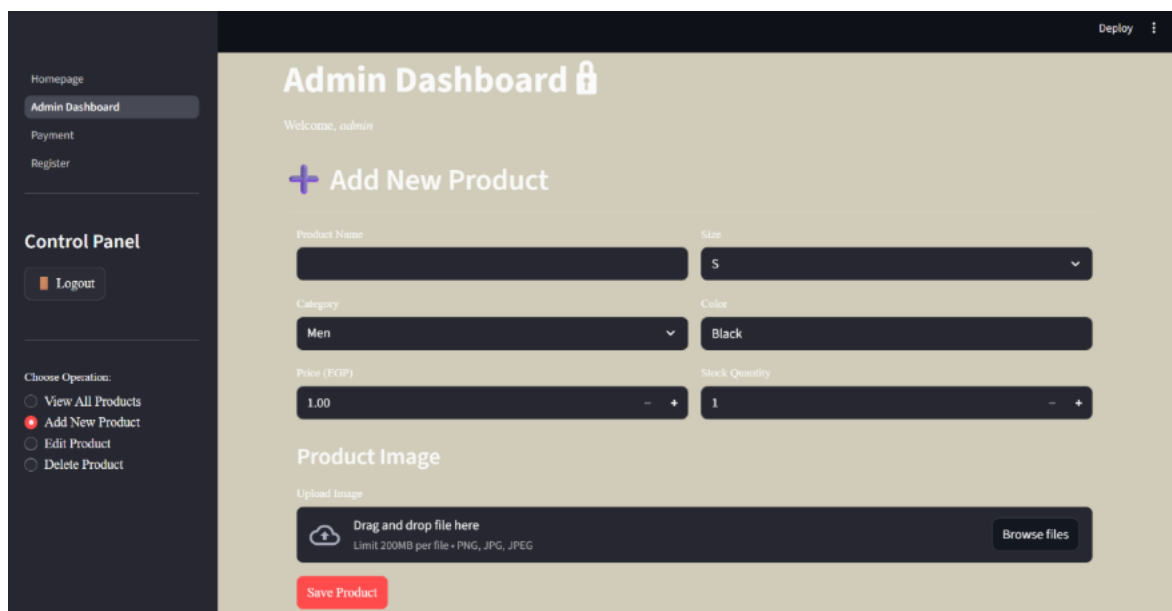
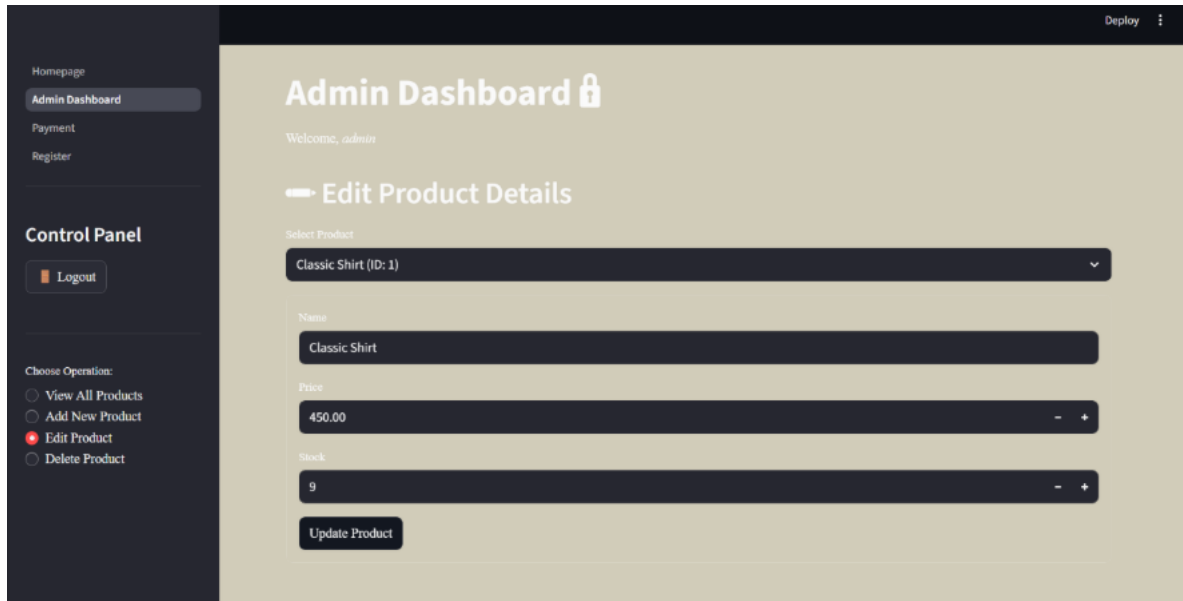


Figure 12. Visualization of Add new product Page

When Admin click Edit product he can edit the details of each product.



The screenshot shows the 'Admin Dashboard' interface. On the left is a dark sidebar with navigation links: 'Homepage', 'Admin Dashboard' (highlighted), 'Payment', and 'Register'. Below these is a 'Control Panel' with a 'Logout' button. At the bottom of the sidebar is a 'Choose Operation:' section with four radio buttons: 'View All Products', 'Add New Product', 'Edit Product' (which is selected), and 'Delete Product'. The main content area has a header 'Admin Dashboard' with a lock icon and a welcome message 'Welcome, admin'. The title of the page is 'Edit Product Details'. It features a 'Select Product' dropdown menu currently showing 'Classic Shirt (ID: 1)'. Below this are input fields for 'Name' (containing 'Classic Shirt'), 'Price' (containing '450.00'), and 'Stock' (containing '9'). Each of these fields has minus and plus icons for adjustment. At the bottom of the form is an 'Update Product' button.

Figure 13. Visualization of Edit product Page

When users click About us the details of our Brand will be shown.

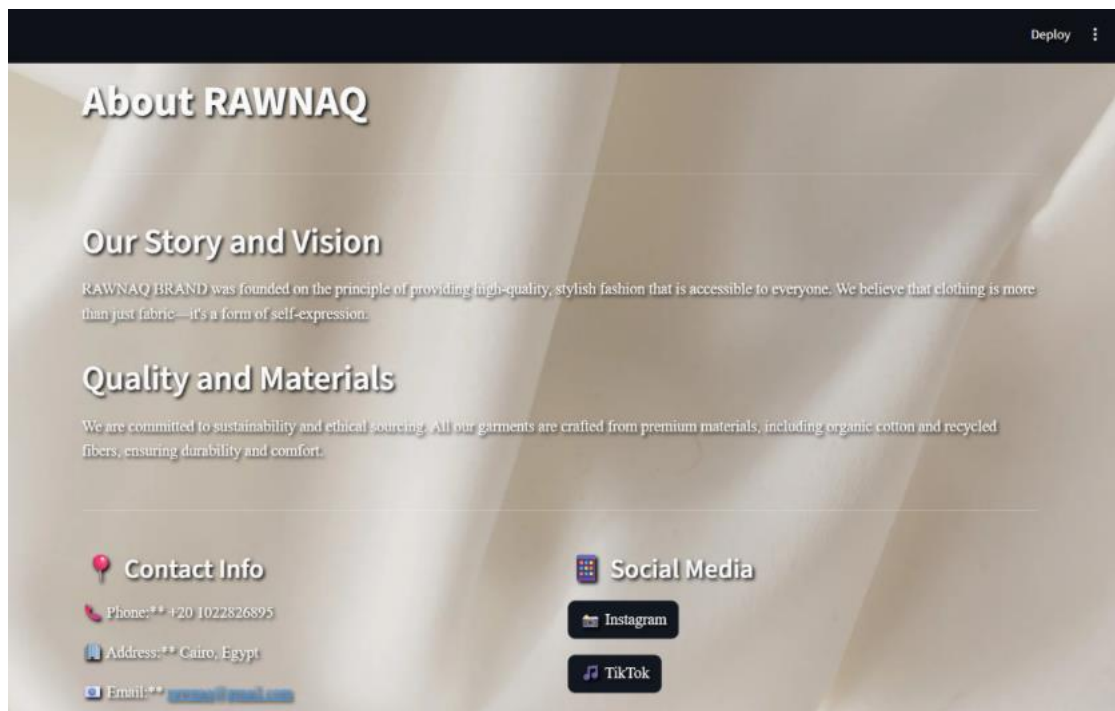


Figure 14. Visualization of Events Page